

ORIGINAL ARTICLE

Exploring ROS Proto-Oncogen 1 (ROS-1), a receptor tyrosine kinase expression in cervical cancer: A clinicopathological analysis

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ABSTRACT

Background: Cervical cancer remains a leading cause of cancer-related mortality among women, with persistent HPV infection as the primary risk factor. Despite advancements in prevention and treatment, diagnosis often occurs at later stages. The exploration of biomarkers in cervical cancer may offer valuable insights into disease progression and potential therapeutic targets. However, the role of biomarkers like ROS1 remains inadequately studied, warranting further investigation to better understand its therapeutic potential.

Methods: This observational cross-sectional study involved 66 cervical cancer patients with a mean age of 47.14 ± 8.58 years. ROS1 expression was assessed using immunohistochemistry (IHC) on biopsy samples. Clinicopathological data, including cancer stage, histology, differentiation grade, and lymphovascular space invasion (LVSI), were collected. Statistical analysis was performed using the Chi-Square test and Mann-Whitney test to evaluate correlation between ROS1 expression and clinical factors.

Results: ROS1 positivity was observed in 56.1% of patients. Significant associations were found between ROS1 expression and parity ($p = 0.040$), clinical stage ($p = 0.028$), and histological subtype ($p = 0.007$). Higher expression was notably linked to advanced stages and adenocarcinoma cases. Mann-Whitney analysis revealed a significant difference in ROS1 expression between squamous and non-squamous histologies ($p = 0.0007$).



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Conclusion: ROS1 expression is significantly correlated with disease progression in cervical cancer, particularly in adenocarcinoma subtypes. These findings suggest that ROS1 could serve as a valuable prognostic biomarker and a potential target for precision therapies in cervical cancer. Further studies are warranted to validate this and explore one as a therapeutic target. (www.actabiomedica.it)

Key words: cervical cancer, ROS1 expression, immunohistochemistry

Introduction

Cervical cancer remains a leading cause of cancer-related mortality among women globally. The World Health Organization (WHO) ranks it as the fourth most common cancer in women, causing over 300,000 deaths each year (1). In Indonesia, cervical cancer is the second most common cancer after breast cancer, with 32,469 new cases and 18,279 deaths reported in 2020 (2). Persistent infection with high-risk Human Papillomavirus (HPV), primarily transmitted through sexual contact, is linked to approximately 99% of cervical cancer cases (3). Additional risk factors include smoking, immunosuppression, multiple sexual partners, and insufficient screening (4). The pathogenesis of cervical cancer begins with persistent HPV infection, leading to cellular lesions in the cervical transformation zone. While some lesions may regress, others progress through stages of cervical intraepithelial neoplasia (CIN I to III), eventually advancing to invasive cancer (5). HPV oncoproteins E6 and E7 disrupt tumor suppressor proteins like p53, impairing cell cycle checkpoints, promoting uncontrolled proliferation, and contributing to genomic instability and cancer progression (6). Cervical cancer is often asymptomatic in its early stages, emphasizing the need for regular screening to enable early detection. As the disease progresses, symptoms such as abnormal vaginal bleeding (especially after intercourse) and unusual vaginal discharge may appear. Histologically, cervical cancer consists primarily of two subtypes: squamous cell carcinoma (SCC), which accounts for approximately 70% of cases, and adenocarcinoma, which represents about 25% (7). The distribution of these subtypes varies

regionally, with SCC predominating in low-income countries, while the incidence of adenocarcinoma has increased in high-income nations (8). The survival outcomes of patients with cervical cancer are strongly influenced by the stage at diagnosis and histological subtype. Patients diagnosed at advanced stages generally exhibit lower survival rates compared to those diagnosed earlier (9). Additionally, histological subtype plays a crucial role in prognosis, with adenocarcinoma generally associated with poorer survival outcomes compared to SCC (10). These factors underscore the importance of early detection and accurate histological assessment in improving patient prognosis. Molecular diagnostics are increasingly important for understanding the genetic mechanisms driving cervical cancer. Among the key oncogenes implicated in the disease, ROS1 (Receptor Tyrosine Kinase Proto-Oncogene) has garnered significant attention. Located on chromosome 6q22, ROS1 encodes a receptor tyrosine kinase that regulates cellular processes such as differentiation, proliferation, and survival (11). In its normal form, ROS1 acts as a receptor for growth factors, activating downstream signaling pathways, including JAK/STAT, PI3K/AKT, and MAPK/ERK, essential for normal cell functioning (12). Genomic alterations, such as ROS1 gene fusions, lead to constitutive activation of ROS1 kinase, enhancing cell proliferation, survival, and migration (13). ROS1 fusions have been extensively studied in non-small cell lung cancer (NSCLC), where they are commonly found in younger, non-smoking patients with adenocarcinoma (14). However, research on ROS1 in cervical cancer remains limited. Preliminary studies suggest that ROS1 may serve as a prognostic biomarker, particularly in cervical

adenocarcinoma, with its expression correlating with disease progression (15). A recent study by Machida et al. (2023) further supports this, indicating that ROS1 gene rearrangements may be associated with the prognosis of cervical adenocarcinoma (16). Given these gaps in the literature, the aim of this study is to investigate ROS1 expression in cervical cancer, with a focus on its association with key clinical and pathological factors such as cancer stage, histological subtype, grade of differentiation, and lymphovascular space invasion (LVSI). Furthermore, this study seeks to explore the potential of ROS1 as a therapeutic target, which could lead to more effective treatment strategies, particularly for patients with advanced or high-risk cervical cancer (17).

Methods

This observational analytical study utilized a cross-sectional design to investigate the expression of ROS1 receptor tyrosine kinase in cervical cancer and its association with various clinicopathological factors. The study was conducted at Hasanuddin University Teaching Hospital in Makassar, Indonesia, beginning in January 2025. A total of 66 female patients diagnosed with cervical cancer were included in the study.

Inclusion and exclusion criteria

Eligible participants were female patients aged 18 to 70 years with a confirmed diagnosis of cervical cancer (stages I-IV). Patients were excluded if they had other malignancies, autoimmune disorders, HIV/AIDS, or if they had previously received chemotherapy or radiotherapy.

Data collection

Patient demographics, cancer stage, histological subtype, lymphovascular space invasion (LVSI), tumor size, and hydronephrosis were collected from medical records. ROS1 expression was evaluated using immunohistochemistry (IHC) on biopsy tissue samples, enabling the assessment of ROS1 protein levels.

Statistical analysis

Statistical analyses were conducted using both univariate and bivariate methods. Descriptive statistics summarized the demographic and clinical characteristics of the study population. The Mann-Whitney U test was applied to compare ROS1 expression levels across groups with non-normally distributed data. The Chi-Square test was used to examine associations between categorical variables.

Ethical considerations

Ethical approval for the study was granted by the Biomedical Research Ethics Committee at Hasanuddin University. All participants provided written informed consent prior to their inclusion, and patient confidentiality was strictly maintained throughout the study.

Result

A total of 66 patients with cervical cancer were included in the study, with a mean age of 47.14 ± 8.58 years. The largest age group consisted of patients aged 45–54 years (43.9%), followed by those aged <45 years (34.8%) and ≥ 55 years (21.2%). The histological analysis revealed that SCC was the predominant subtype (87.8%), while adenocarcinoma/adenosquamous carcinoma accounted for 12.2%. Immunohistochemical analysis showed that 56.1% of patients were ROS1-positive, with the remaining 43.9% being ROS1-negative. The distribution of these findings, along with other clinical and pathological variables such as tumor size, degree of differentiation, hydronephrosis, and contraceptive use, is summarized in Figure 1.

As shown in Figure 2. below, a significant association was found between parity and ROS1 expression ($p = 0.040$), with a higher proportion of multiparous patients exhibiting ROS1 positivity. No significant associations were observed between age, sexual partner history, or contraceptive use and ROS1 expression ($p = 0.894$, $p = 0.575$, $p = 0.639$, respectively).

Bivariate analysis revealed a significant association between clinical stage and ROS1 expression ($p = 0.028$), with a higher percentage of ROS1-positive

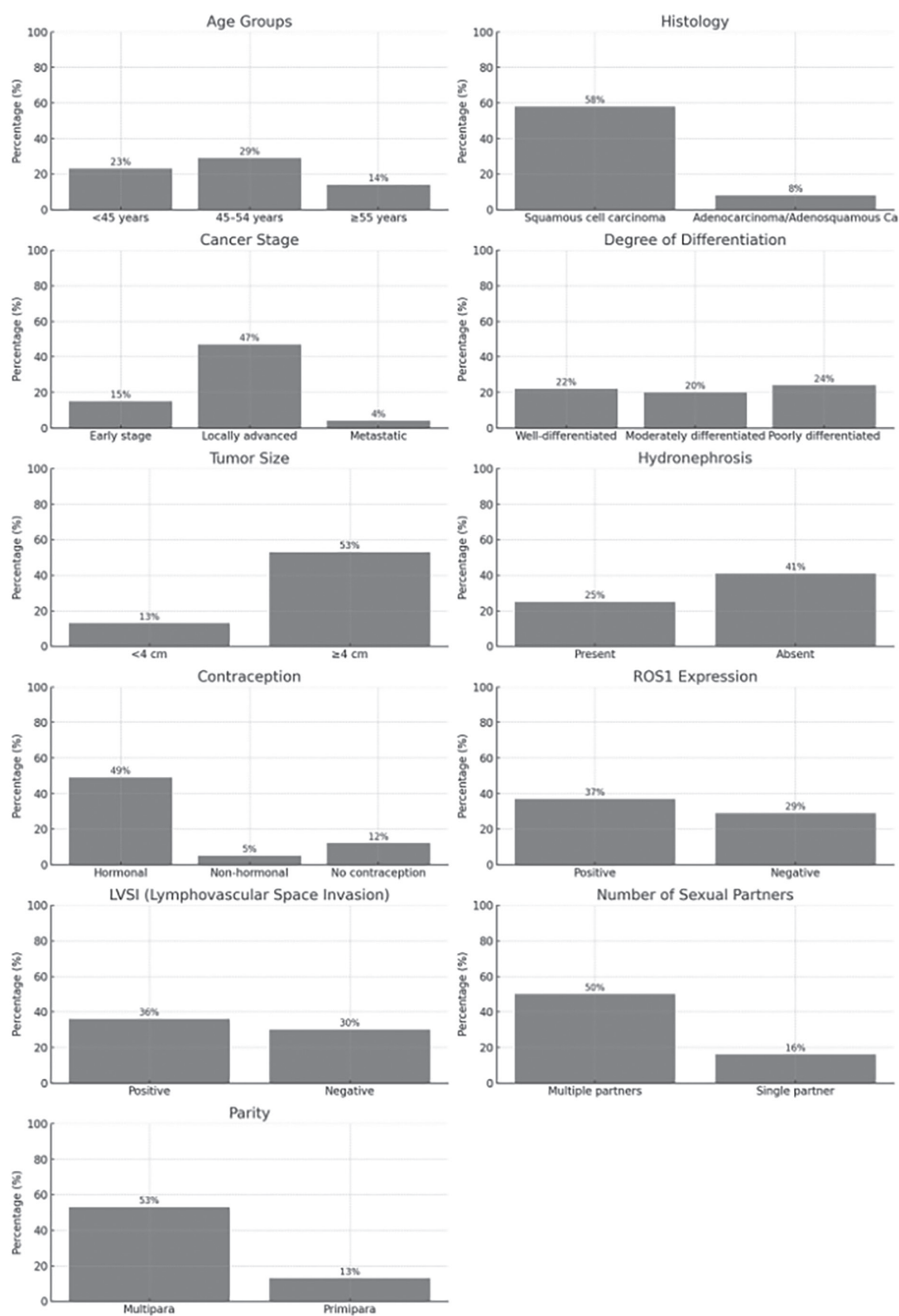


Figure 1. Sample Characteristics

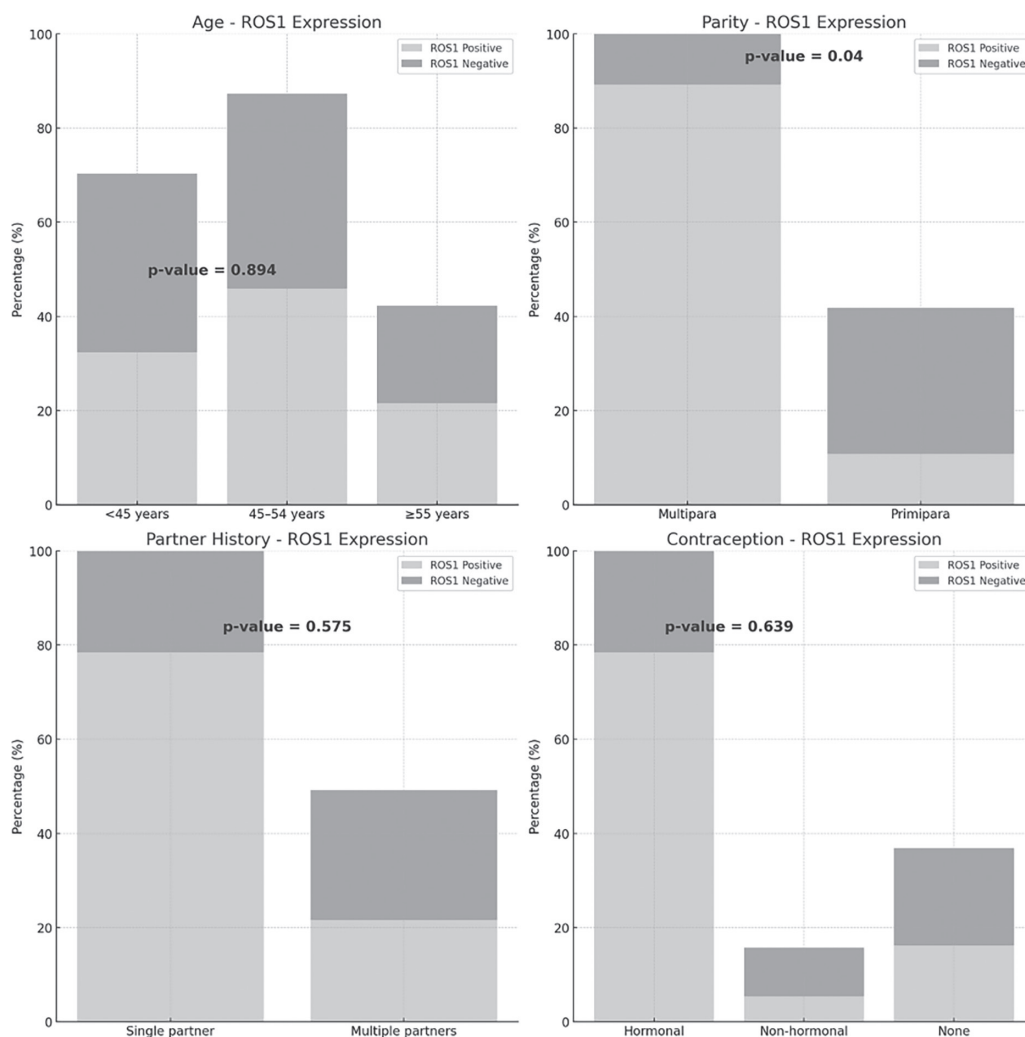


Figure 2. ROS1 and its Association with Host Factors

cases in the locally advanced group (67.6%). Additionally, histological type showed a significant relationship with ROS1 expression ($p=0.007$), as all adenocarcinoma /adenosquamous carcinoma cases were ROS1-positive, while 78.4% of squamous cell carcinoma cases exhibited ROS1 positivity. This is summarized in Figure 3.

ROS1 expression was significantly associated with cancer stage and histological subtype. The Kruskal-Wallis test revealed a significant difference in ROS1 expression across cancer stages ($p = 0.041$). The metastatic stage exhibited the highest median H-Score (180), followed by locally advanced (110), and early stage (80). The Mann-Whitney U test showed

a significant difference in ROS1 expression between squamous cell carcinoma and adenocarcinoma/adenosquamous carcinoma ($p = 0.0007$), with the latter group displaying a higher median H-Score (240 vs. 95) as shown in Figure 4.

In this study, Adenocarcinoma with positive ROS1 expression was analyzed using immunohistochemistry (IHC). The expression intensity of ROS1 was evaluated by assigning an H-Score, which quantifies staining intensity and the percentage of positive cells. The H-Score in this case ranged from 0 to 270, with 270 representing the highest intensity of ROS1 expression. The following sections detail the findings from this analysis at various magnifications.

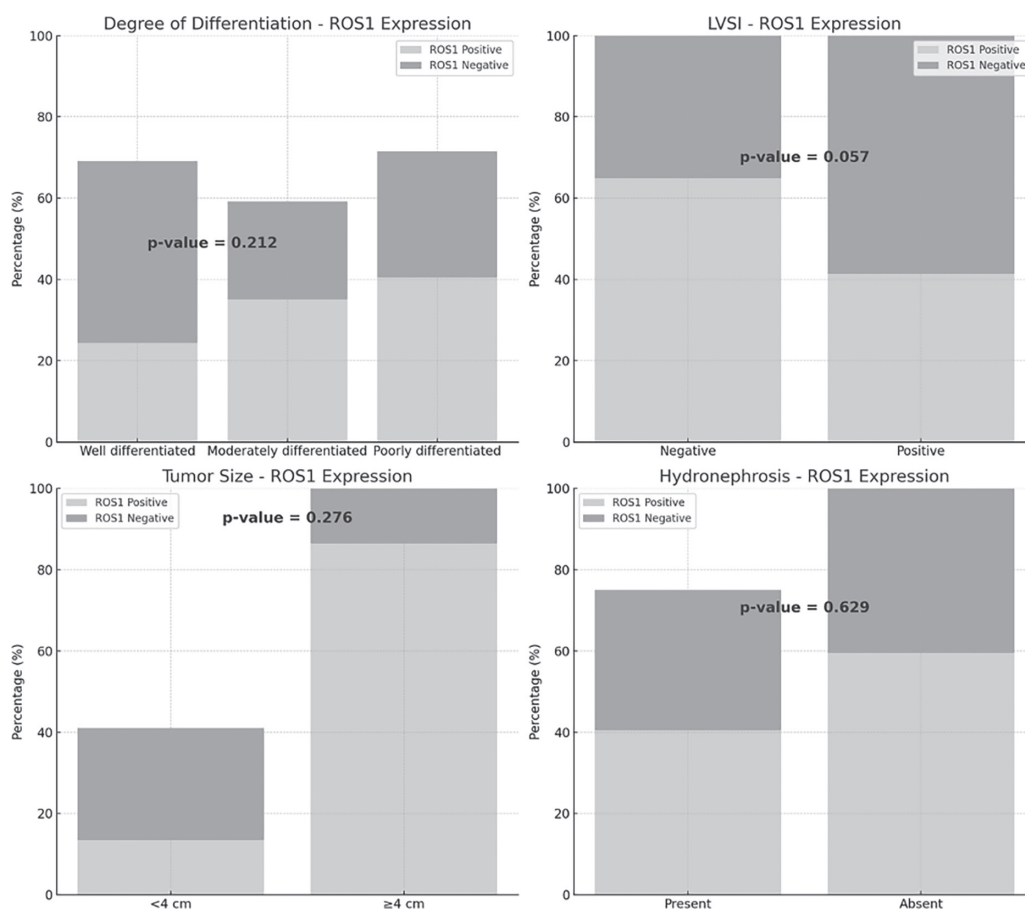


Figure 3. ROS1 Expression and Clinicopathology Factors

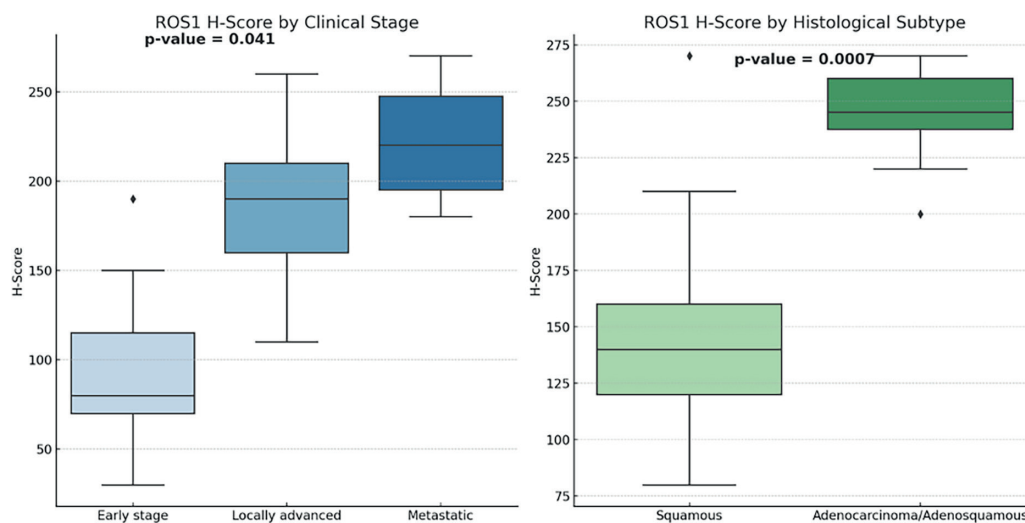


Figure 4. Analysis of ROS1 H-Score Values Across Different Stages and Histological Subtypes

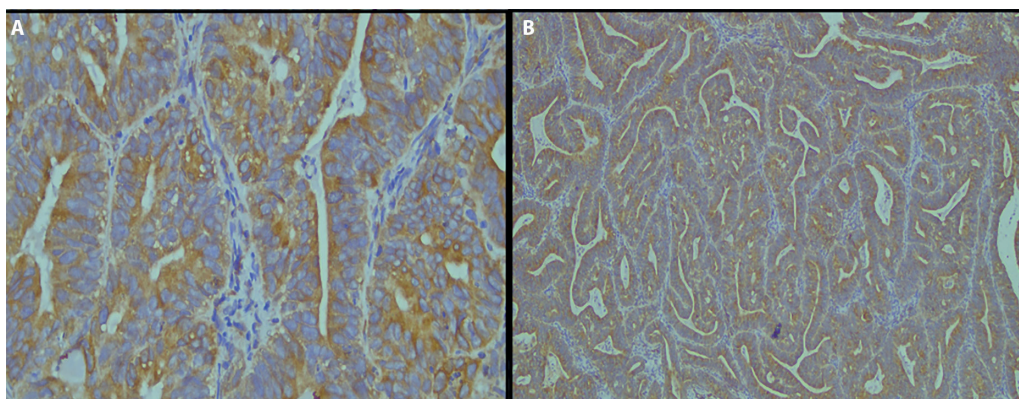


Figure 5. Immunohistochemical staining of ROS1 expression in adenocarcinoma of the cervix.

A) 40x magnification: Strong cytoplasmic and membranous staining of ROS1 in adenocarcinoma cells, with an H-Score of 270, indicating high expression intensity. The tumor cells exhibit pronounced brown staining, confirming the presence of ROS1.

B) 10x magnification: Overview of the ROS1 expression across the tumor, showing consistent membranous and cytoplasmic staining throughout. The H-Score of 270 reflects widespread ROS1 positivity, consistent with the high expression observed at higher magnification.

Figure 5.A (40x Magnification), At a 40x magnification, Adenocarcinoma cells exhibited strong cytoplasmic and membranous ROS1 staining, indicative of robust ROS1 expression. The H-Score for this sample was 270, indicating an exceptionally high level of ROS1 positivity. The intense brown staining observed in the tumor cells is characteristic of ROS1 positivity, highlighting the extensive expression within the cell membranes and cytoplasm. The staining intensity was consistent across the tumor, reinforcing the idea that ROS1 fusion is a prominent feature of this tumor type.

Figure 5.B (10x Magnification): At 10x magnification, the overall distribution of ROS1 expression across the tumor became evident, though individual cellular details were less distinguishable. The H-Score of 270 remained consistent across this section, confirming that ROS1 expression is not only concentrated in specific areas but is widespread throughout the tumor. The membranous and cytoplasmic staining observed at this magnification correlates with ROS1

activation and fusion in adenocarcinoma cells, suggesting a potential molecular pathway driving tumor growth and progression. The high ROS1 expression observed in adenocarcinoma underscores its potential as a biomarker for this histological subtype. Given the strong correlation between ROS1 positivity and tumor aggression, the findings suggest that ROS1 could serve as both a prognostic biomarker and a therapeutic target. The consistent expression across varying magnifications further emphasizes the potential clinical relevance of ROS1 as a targetable marker for adenocarcinoma in cervical cancer, offering insight into new therapeutic avenues for patients with this subtype.

Discussion

ROS1 expression and clinicopathological factors

This study demonstrates a significant association between ROS1 expression and cancer stage, particularly in advanced stages of cervical cancer. Patients diagnosed at locally advanced and metastatic stages exhibited significantly higher levels of ROS1 expression, suggesting that ROS1 plays a crucial role in tumor progression and aggressiveness. This is consistent with findings in other cancers, where ROS1 activation is associated with increased tumor proliferation, survival, and metastasis through signaling pathways like MAPK/ERK and PI3K/AKT (1,2). The higher ROS1

expression observed in advanced stages highlights its potential as a prognostic biomarker for cervical cancer, aiding in the identification of patients at higher risk for disease progression.

Interestingly, although metastatic cases generally exhibited high ROS1 expression, one case in the metastatic group showed negative ROS1, emphasizing the heterogeneity of cervical cancer. This variability suggests that ROS1 activation is not the sole determinant of tumor aggressiveness, and other molecular mechanisms may contribute to cancer progression, particularly in metastatic disease. These findings reinforce the complexity of cervical cancer biology and underscore the need for personalized treatment strategies, where molecular profiling could help guide therapy decisions.

In this study, ROS1 expression was also significantly associated with histological subtype ($p = 0.007$). All cases of adenocarcinoma and adenosquamous carcinoma were ROS1-positive, while squamous cell carcinoma (SCC) exhibited a more variable expression pattern. This finding aligns with previous research suggesting that ROS1 may serve as a prognostic marker for adenocarcinomas, but not necessarily for SCC (3). Adenocarcinoma, often linked to receptor tyrosine kinase (RTK) pathway activation (e.g., MAPK/ERK and PI3K/AKT), may rely more on ROS1 activation for tumor growth and invasiveness (4). In contrast, SCC is typically driven by HPV-related alterations in the p53 and Rb pathways, which do not commonly involve ROS1 activation (5). These findings highlight the importance of considering histological subtype when evaluating ROS1 as a biomarker for cervical cancer.

In addition to cancer stage and histology, ROS1 expression was significantly associated with tumor size ($p = 0.015$). Larger tumors (>4 cm) exhibited significantly higher ROS1 expression, further supporting the idea that ROS1 is involved in tumor aggressiveness. Larger tumors are typically indicative of advanced disease and poor prognosis, and our findings suggest that ROS1 expression could serve as an early indicator of aggressive disease. This association with tumor size also suggests that ROS1 may be a valuable target for therapies aimed at locally advanced cervical cancer.

ROS1 expression and host factors

The study also explored the relationship between ROS1 expression and host factors, including age, parity, and sexual history. A significant association was found between ROS1 positivity and multiparity ($p = 0.040$), suggesting that reproductive history may play a key role in the molecular landscape of cervical cancer. Multiparity increases the risk of cervical cancer through mechanisms such as cervical trauma during childbirth, hormonal changes, and prolonged HPV exposure over time (6,7). Our study demonstrated that 89.2% of ROS1-positive cases were found in multiparous women, highlighting the potential impact of reproductive history on ROS1 activation. As a receptor tyrosine kinase, ROS1 is involved in regulating cell proliferation and survival, and its activation could contribute to tumor development in multiparous women by promoting these processes (8). This finding suggests that multiparity may enhance the biological aggressiveness of cervical cancer by activating ROS1 signaling pathways. Interestingly, age did not show a significant correlation with ROS1 expression in our study, which is consistent with the understanding that ROS1 activation is more closely related to the intrinsic tumor biology than to the patient's age. Previous studies have emphasized that ROS1 activation is driven by tumor-specific factors, such as histology and genomic alterations, rather than by patient demographics (9,10). Thus, ROS1 may act as a molecular marker reflecting the biological behavior of the tumor rather than external factors like age. Additionally, sexual history and the number of sexual partners did not correlate significantly with ROS1 expression ($p = 0.575$). This suggests that while sexual history is a recognized risk factor for HPV infection, which is crucial for the initiation of cervical carcinogenesis, ROS1 activation may be independent of sexual behavior. Our results support the concept that HPV persistence and the immune response play a more critical role in disease progression, with ROS1 activation likely driven by intratumoral dynamics rather than external exposures alone (11,12). The fact that many ROS1-positive patients in this study had a history of single-partner sexual contact reinforces the idea that HPV transmission can occur through long-term exposure, and suggests that ROS1

expression is influenced more by host factors such as immune status than by the number of sexual partners.

ROS1 expression across cancer stages and histological subtypes

This study demonstrated a significant association between ROS1 expression and both cancer stage and histological subtype, with the H-score providing a quantitative assessment of ROS1 activation. The Kruskal-Wallis test revealed a significant difference in ROS1 expression across different cancer stages ($p = 0.041$). Specifically, the metastatic stage exhibited the highest median H-score (180), followed by locally advanced stage (110), and early stage (80), indicating that ROS1 expression increases as the disease progresses from early to metastatic stages. This suggests that ROS1 plays a significant role in tumor progression and aggressiveness, with higher H-scores reflecting more aggressive tumor behavior. ROS1 activation is known to drive cell proliferation, survival, and migration through critical signaling pathways such as MAPK/ERK and PI3K/AKT (1,2). The higher H-score in metastatic tumors is consistent with the role of ROS1 in facilitating tumor spread and suggests its potential as a prognostic biomarker for advanced-stage cervical cancer.

In addition to cancer stage, ROS1 expression was significantly associated with histological subtype ($p = 0.0007$). The H-score revealed that adenocarcinoma and adenosquamous carcinoma cases had significantly higher ROS1 expression compared to squamous cell carcinoma (SCC). Specifically, adenocarcinoma/adenosquamous carcinoma had a median H-score of 240, while SCC had a much lower median H-score of 95, indicating a marked difference in ROS1 expression between these histological subtypes. This suggests that ROS1 activation plays a more significant role in driving the aggressiveness of adenocarcinoma, which is often associated with receptor tyrosine kinase (RTK) pathway activation, such as MAPK/ERK and PI3K/AKT signaling (3,4). In contrast, SCC, typically driven by HPV-induced alterations in the p53 and Rb pathways, does not generally involve ROS1 activation (5). These results emphasize the need to consider histological subtype when evaluating ROS1 expression as a potential biomarker for cervical cancer. The higher

H-scores in adenocarcinoma and adenosquamous carcinoma also suggest the potential for ROS1-targeted therapies, which could provide a promising treatment option for patients with ROS1-positive adenocarcinomas, especially those with advanced disease.

Interplay between HPV infection and ROS1 expression in cervical cancer

Interestingly, persistent high-risk HPV infection drives cervical carcinogenesis through the viral oncoproteins E6 and E7, which inactivate tumor suppressors such as p53 and Rb, disrupt cell cycle control, and promote genomic instability, mechanisms that set the stage for additional molecular alterations during tumor progression (3). While there is no direct evidence showing that high-risk HPV suppresses ROS1 expression specifically, the complex interplay between HPV-induced dysregulation and other oncogenic pathways could influence how receptor tyrosine kinases like ROS1 are expressed or activated as tumors evolve. In support of the relevance of ROS1 in cervical cancer, recent exploratory analyses suggest that ROS1 gene alterations may serve as a prognostic biomarker in cervical adenocarcinoma, particularly in more aggressive disease subtypes (16). This suggests that ROS1 could be a key player in the progression and treatment resistance of cervical adenocarcinoma, especially in patients who do not respond well to conventional therapies targeting HPV-related mechanisms.

Implications for clinical practice

The findings of this study suggest important clinical implications for the management of cervical cancer, particularly regarding the role of ROS1 expression. The significant association between ROS1 activation and advanced cancer stages, as well as adenocarcinoma histology, indicates that ROS1 could be a valuable prognostic biomarker for predicting disease progression and aggressiveness. ROS1 expression may serve as an early indicator of tumor invasiveness and metastatic potential, especially in patients with locally advanced or metastatic disease. Incorporating ROS1 assessment into clinical practice could help identify high-risk patients who are likely to benefit from targeted therapies aimed at inhibiting ROS1 signaling pathways. The

findings also underscore the need for a more personalized treatment approach in cervical cancer, particularly for adenocarcinoma and adenosquamous carcinoma, which exhibit higher ROS1 expression compared to SCC. Given that adenocarcinoma tends to be more aggressive and less responsive to traditional therapies, ROS1-targeted therapies could offer a promising new treatment option for ROS1-positive tumors. Further clinical studies are needed to validate the efficacy of ROS1 inhibitors in cervical adenocarcinoma and adenosquamous carcinoma, especially in patients with advanced disease. In clinical practice, HPV genotyping combined with ROS1 testing could provide a more comprehensive understanding of tumor biology, allowing for better stratification of patients based on tumor aggressiveness and metastatic risk. Targeted therapies directed against ROS1 could improve treatment outcomes for patients with ROS1-positive cervical cancer, particularly those with resistant or advanced-stage disease.

Strengths and limitations

This study represents one of the first investigations of ROS1 expression in cervical cancer in Indonesia, particularly using immunohistochemistry (IHC) to evaluate ROS1 as a prognostic biomarker. Given the limited molecular data available in the region, this research adds significant value to the local body of knowledge, which often focuses more on classical risk factors and clinicopathological characteristics. Moreover, the study uses quantitative methods to assess ROS1 expression through the H-score, offering a more nuanced understanding of the intensity of ROS1 activation compared to traditional positive/negative classifications. This approach allows for a detailed comparison across different histological subtypes, which enhances our understanding of the molecular heterogeneity of cervical cancer. In addition to these methodological strengths, this study has made a novel contribution to the field by demonstrating the potential of ROS1 as a prognostic biomarker in cervical adenocarcinoma, a subtype that is typically associated with poor prognosis and aggressive disease. Unlike previous research that has primarily focused on HPV-related molecular mechanisms, our study underscores the potential for ROS1-targeted

therapies to improve treatment outcomes for ROS1-positive cervical cancers. This exploration of ROS1 in cervical cancer highlights its therapeutic relevance and positions ROS1 as a potential target for personalized treatment strategies in the future. The study has several limitations that must be considered when interpreting the findings. One major limitation is the small sample size, particularly for adenocarcinoma and adenosquamous carcinoma cases, which limits the ability to generalize the results to the broader population. The small sample size also reduces the statistical power, especially in subgroup analyses. Another limitation is the lack of long-term follow-up, which prevents assessment of ROS1 expression's prognostic value in relation to treatment outcomes, disease recurrence, and patient survival. As ROS1 expression was not correlated with clinical outcomes, further studies with extended follow-up periods and larger sample sizes are necessary to evaluate the role of ROS1 as a predictive biomarker for response to treatment and risk of recurrence in cervical cancer.

Conclusion

This study highlights the significant role of ROS1 expression in cervical cancer, particularly in the adenocarcinoma and adenosquamous carcinoma subtype. Findings indicate that higher ROS1 expression is associated with advanced cancer stages and adenocarcinoma histology, suggesting that ROS1 plays a crucial role in the progression of cervical cancer. Moreover, the study emphasizes the importance of considering ROS1 as a potential therapeutic target, particularly for patients with adenocarcinoma, a subtype that is often less responsive to conventional treatments. The significant association between ROS1 and clinical stage, along with its widespread expression in non-squamous histologies, underscores its potential as a prognostic biomarker.

Ethical Approval: This study was approved by the Ethics Committee of the Faculty of Medicine, Hasanuddin University (No.476/KEPK/UNHAS/2024) and conducted in accordance with the principles of the Declaration of Helsinki.

Conflict of Interest: Each author declares that he or she has no commercial associations (e.g. consultancies, stock

ownership, equity interests, patent/licensing arrangements, etc.) that might give rise to a conflict of interest with respect to the submitted article.

Authors' Contributions: NHH: Conceptualization, Study design, Writing – original draft; NUP: Data curation, Clinical supervision, Writing – review and editing; LML, SA: Conceptualization, Study design, Supervision, Analysis, Data interpretation, Literature search; RM: Immunohistochemical examinations, Pathological data validation; AAZ: Statistical analysis, Data interpretation, Writing – review and editing. All authors read and approved the final version of the manuscript and agree to be accountable for all aspects of the work.

Declaration on the Use of AI: No artificial intelligence tools were used in the writing or editing of this manuscript.

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